Existence and Smoothness of Elementary Weak Solutions of the Navier–Stokes Equations: Elementary Partial Differential Equations Theory

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Abstract

We construct solutions depending on external forces and not on prescribed initial values. Our solutions are unique, smooth, analytic, and time global for small data. The method is similar to the Banach fixed-point theorem, but we do not assume that the image of the domain is contained in the domain itself.

1 Introduction

The existence of solutions is actually known. For example, there are the Fujita–Kato theory and the Shibata theory; see Kato [4], Zhang [12], Charve–Danchin [2], and Shibata–Murata [6].

Semigroup theory or a priori estimates appear in these theories, but they are not elementary. Semigroup theory or a priori estimates for contraction mappings are not used in the proof of the existence of Leray–Hopf weak solutions, for example in Shibagaki–Rikimaru [7], but those arguments are not elementary either.

We define new weak solutions with uniqueness and smoothness without using semigroup theory or a priori estimates. We apply the local solvability of linear partial differential operators with constant coefficients. The basic policy is to take

$$L = \partial_t - \Delta$$

as the heat operator in the initial value problem for the Navier–Stokes equations

$$\partial_t u - \Delta u = f - \nabla p - (u \cdot \nabla)u, \tag{N-S}$$

$$\operatorname{div} u = 0, \tag{σ}$$

$$u(0,x) = a(x), \tag{A}$$

together with the decay property

$$\lim_{t,\,|x|\to\infty} \partial^\alpha u(t,x) = 0, \tag{S}$$

and continuity of the maps

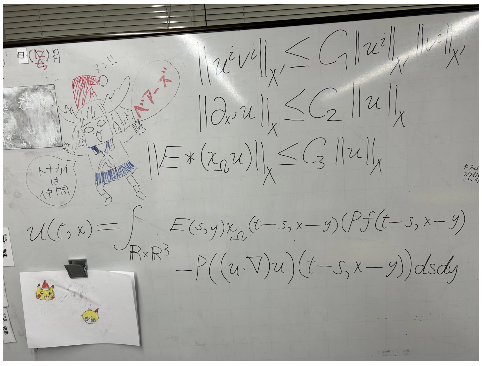
$$f \longmapsto u, \qquad a \longmapsto u. \tag{C}$$

The uniqueness follows without imposing boundary values for u. Our solutions decrease rapidly, satisfy an energy inequality, and are analytic. We do not use long or complicated calculations, semigroup theory, a priori estimates, or a theory of evolution equations.

We apply the local solvability of linear partial differential operators with constant coefficients [3, 8], eliminate the pressure by the Helmholtz projection P, and solve

$$\partial_t u - \Delta u = Pf - P\big((u_n \cdot \nabla)u_n\big),$$

$$\partial_t u - \Delta u = Pf - P\big((u \cdot \nabla)u\big)$$



in a Sobolev-type space constructed below.

Let

$$\Omega \subset \mathbb{R} \times \mathbb{R}^3$$

be a bounded domain with Lipschitz boundary such that

$$(0,0) \in \Omega.$$

A typical example is the cylinder

$$\Omega = I \times \Omega' = (-T,T) \times B^3(0,R).$$

We handle the variables t, x simultaneously. Cylinder domains are Lipschitz domains, so the Sobolev embedding theorem can be used on Ω, and the Helmholtz decomposition can be used on the spatial domain Ω'.

2 Definition

For convenience, the index of a component of a vector is written as an upper index. The words "function space" and "space" mean linear topological spaces of functions or distributions. Except for the pressure p, functions are $\mathbb{R}^3$-valued unless otherwise stated. We use the same symbols for spaces of real-valued and $\mathbb{R}^3$-valued functions when no confusion occurs.

Let $|\Omega|$ be the Lebesgue measure of Ω, and let χ_Ω denote the characteristic function of Ω. For

$$m > \max\left\{\frac{4}{1}, \frac{4}{2}\right\} = 4, \qquad p = 1, 2,$$

define

$$V^{m,p}(\Omega) = \left\{u = (u^1, u^2, u^3) : u^i \in C^\infty(\Omega),\ \left\|u^i\right\|_{W^{m,p}(\Omega)} < \infty\right\},$$

and

$$V^{m,p}_\sigma(\Omega) = \left\{u \in V^{m,p}(\Omega) : \operatorname{div} u = 0\right\}.$$

Let $W^{m,p}(\Omega)$ and $W^{m,p}_\sigma(\Omega)$ be the Sobolev spaces obtained as the completions of these spaces.

Let $\mathcal{D}(\Omega) = C_0^\infty(\Omega)$, and let

$$\mathcal{D}_\sigma(\Omega) = \left\{\varphi \in \mathcal{D}(\Omega)^3 : \operatorname{div}_x \varphi = 0\right\}.$$

Let

$$P : L^2(\Omega) \longrightarrow L^2_\sigma(\Omega)$$

be the Helmholtz projection.

For vector fields w, φ, put

$$\langle w, \varphi\rangle = (w, \varphi)_{L^2(\Omega)} = \int_\Omega w(t,x)\cdot\varphi(t,x)\,dt\,dx = \int_\Omega \sum_{i=1}^3 w^i(t,x)\varphi^i(t,x)\,dt\,dx.$$

If X, Y are Banach spaces continuously contained in a Hausdorff linear space Z, then $X \cap Y$ is a Banach space with either of the equivalent norms

$$\|u\|_X + \|u\|_Y, \qquad \max\{\|u\|_X, \|u\|_Y\}.$$

Indeed,

$$\max\{\|u\|_X, \|u\|_Y\} \le \|u\|_X + \|u\|_Y \le 2\max\{\|u\|_X, \|u\|_Y\}.$$

Put

$$\mathcal{X}_\sigma = \bigcap_{m\ge 5}\left(W^{m,1}_\sigma(\Omega) \cap W^{m,2}_\sigma(\Omega)\right),$$

and

$$\mathcal{X}' = \bigcap_{m\geq 5} \left(W^{m,1}(\Omega) \cap W^{m,2}(\Omega)\right).$$

For suitable positive constants $c'_{k,m}$, define

$$\|u\|_X = \sum_{m=5}^{\infty}\sum_{k=0}^{\infty} \frac{1}{(m!)^4 (c'_{k,m})^2} \|u\|_{W^{m,1}(\Omega)\cap W^{m,2}(\Omega)}$$

for $u \in \mathcal{X}_\sigma$, and similarly

$$\|u\|_{X'} = \sum_{m=5}^{\infty}\sum_{k=0}^{\infty} \frac{1}{(m!)^4 (c'_{k,m})^2} \|u\|_{W^{m,1}(\Omega)\cap W^{m,2}(\Omega)}$$

for $u \in \mathcal{X}'$.

Then put

$$X = \{u \in \mathcal{X}_\sigma : \|u\|_X < \infty\}, \qquad X' = \{u \in \mathcal{X}' : \|u\|_{X'} < \infty\}.$$

For constants $M, C > 0$, define

$$S = \{u \in X : \|u\|_X \leq M, \quad \|\partial_{x^j} u\|_X \leq C\|u\|_X, \ j = 1,2,3\}.$$

Let E be the fundamental solution of the heat operator

$$\partial_t - \Delta.$$

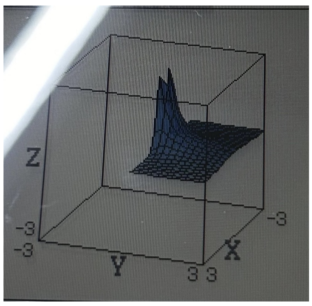
Thus

$$(\partial_t - \Delta)E = \delta(t) \otimes \delta(x)$$

in the sense of distributions, where

$$E(t,x) = \begin{cases} \dfrac{1}{(4\pi t)^{3/2}} \exp\left(-\dfrac{|x|^2}{4t}\right), & t > 0, \\ 0, & t \leq 0. \end{cases}$$

3 Existence of elementary weak solutions



Assumption

Let

$$\Omega \subset \mathbb{R} \times \mathbb{R}^3$$

be a bounded domain with Lipschitz boundary and

$$(0,0) \in \Omega.$$

Let

$$f : \mathbb{R} \times \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

be an external force such that

$$Pf \in S.$$

If $f \neq 0$, assume further that there exists $(0,x) \in \Omega$ such that

$$\int_{\mathbb{R}\times\mathbb{R}^3} E(s,y)(\chi_\Omega Pf)(-s, x-y)\, ds\, dy \neq 0.$$

Let A denote the set of initial values

$$A = \left\{u(0,\cdot) : u \in X,\ u(t,x) = \int_{\mathbb{R}\times\mathbb{R}^3} E(s,y)\big(\chi_\Omega P(f - (u\cdot\nabla)u)\big)(t-s, x-y)\, ds\, dy\right\}.$$

Proposition 3.1 (Local solvability of linear partial differential operators with constant coefficients)**.** *Let L be a linear partial differential operator with constant coefficients on $\mathbb{R}^N$, and let $E \in \mathcal{D}'(\mathbb{R}^N)$ be a fundamental solution:*

$$LE = \delta.$$

For $f \in L^1_{\mathrm{loc}}(\mathbb{R}^N)$ and a bounded domain $\Omega \subset \mathbb{R}^N$, a weak solution of

$$Lu = f \quad \text{in } \Omega$$

is given by

$$u = E * (\chi_\Omega f) \in \mathcal{D}'(\Omega).$$

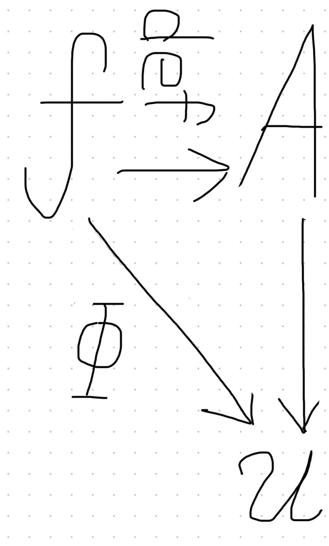
Proof. For every $\varphi \in \mathcal{D}(\Omega)$,

$$\begin{aligned}
\langle L(E * (\chi_\Omega f)), \varphi \rangle &= \pm \langle E * (\chi_\Omega f), L\varphi \rangle \\
&= \pm \langle E(x), \langle \chi_\Omega(y) f(y), L\varphi(x+y) \rangle \rangle \\
&= \pm \langle \chi_\Omega(x) f(x), \langle E(y), L\varphi(x+y) \rangle \rangle \\
&= \langle \chi_\Omega(x) f(x), \langle LE(y), \varphi(x+y) \rangle \rangle \\
&= \langle f, \varphi \rangle .
\end{aligned}$$

□

Theorem 3.2 (Main theorem)**.** *The set A is nonempty. For every $a \in A$, there exist elementary weak solutions u, p of*

$$\begin{aligned}
\partial_t u - \Delta u &= f - \nabla p - (u \cdot \nabla) u, \\
\operatorname{div} u &= 0, \\
u(0, x) &= a(x)
\end{aligned}$$



in Ω, in the sense that

$$u \in X, \qquad p \in L^2_{\mathrm{loc}}(\Omega)$$

up to an additive spatial constant, and

$$\langle \partial_t u + (u \cdot \nabla) u - \Delta u + \nabla p - f, \varphi \rangle = 0$$

for every

$$\varphi \in \mathcal{D}_\sigma(\Omega),$$

while

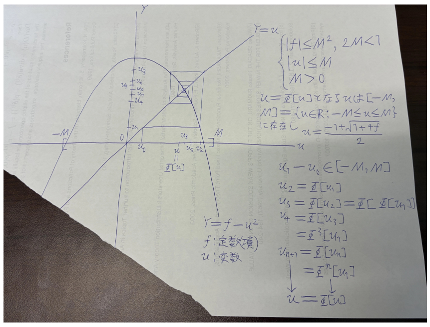
$$\langle \operatorname{div} u, \varphi \rangle = - \sum_{j=1}^{3} \left\langle u^j, \partial_{x^j} \varphi \right\rangle = 0$$

for every $\varphi \in \mathcal{D}(\Omega)$.

For every multi-index

$$\alpha \in \mathbb{Z}^4_{\geq 0},$$

$$\lim_{t,\, |x| \to \infty} \partial^\alpha u(t, x) = 0.$$



If two solutions u, p and v, q satisfy

$$u, v \in S, \qquad u - v \in S,$$

then

$$u = v, \qquad p = q$$

on Ω. The map

$$f \longmapsto u$$

is continuous.

Proposition 3.3 (Smoothness of elementary weak solutions)**.** *The elementary weak solutions* (u, p) *are* C^∞*-functions.*

Proof. Since $m \in \mathbb{N}$ may be taken arbitrarily large, the Sobolev embedding theorem gives, whenever

$$N < m - \frac{4}{p},$$

a continuous embedding

$$W^{m,p}(\Omega) \hookrightarrow C^{N,\alpha}(\Omega), \qquad 0 < \alpha < 1.$$

Hence u has a C^∞-representative. Since

$$\partial_t u + (u \cdot \nabla)u - \Delta u - f = -\nabla p$$

is smooth, p is smooth as well. $\square$

Lemma 3.4. *The normed spaces* X *and* X' *are nontrivial.*

Proof. It is enough to exhibit nonzero elements. For example, suitable restrictions of

$$\operatorname{curl}(e^{x_1}, e^{x_3}, e^{x_2})$$

belong to the divergence-free class, while functions such as

$$e^{x_1}, \qquad \sin x_2$$

give nonzero elements of the corresponding scalar spaces. Thus

$$X \neq \{0\}, \qquad X' \neq \{0\}.$$

$\square$

Lemma 3.5 (Completeness)**.** *The spaces* X *and* X' *are Banach spaces.*

Proof. Let $\{u_n\}$ be a Cauchy sequence in X. Then it is Cauchy in every

$$W^{m,1}_\sigma(\Omega) \cap W^{m,2}_\sigma(\Omega).$$

Since these are Banach spaces, u_n converges in each of them to a common limit u. For every $\varepsilon > 0$, there is N such that for $l, n \geq N$,

$$\|u_l - u_n\|_X < \varepsilon.$$

By Fatou's lemma for counting measure,

$$\begin{aligned} \|u - u_n\|_X &= \sum_{m=5}^{\infty} \sum_{k=0}^{\infty} \frac{1}{(m!)^4 (c'_{k,m})^2} \|u - u_n\|_{W^{m,1} \cap W^{m,2}} \\ &\leq \liminf_{l \to \infty} \sum_{m=5}^{\infty} \sum_{k=0}^{\infty} \frac{1}{(m!)^4 (c'_{k,m})^2} \|u_l - u_n\|_{W^{m,1} \cap W^{m,2}} \\ &\leq \varepsilon. \end{aligned}$$

Therefore,

$$\|u\|_X \leq \|u - u_N\|_X + \|u_N\|_X < \infty,$$

and $u \in X$. The proof for X' is identical. $\square$

Lemma 3.6 (Separation of product)**.** *There exists a constant $C_1 > 0$ such that*

$$\left\|u^i v^i\right\|_{X'} \leq C_1 \left\|u^i\right\|_{X'} \left\|v^i\right\|_{X'}$$

for all $u, v \in X'$.

Proof. For multi-indices $\alpha, \beta \in \mathbb{Z}^4_{\geq 0}$, let $c_{\alpha,\beta}$ denote the binomial coefficients appearing in the Leibniz formula, and set

$$c_\alpha = \sum_{\beta \leq \alpha} c_{\alpha,\beta}.$$

There is a continuous embedding

$$X' \hookrightarrow C^{k,\gamma}(\Omega)$$

for every $k \in \mathbb{N}$, $0 < \gamma < 1$. Thus, for suitable constants $c'_{k,m} > 0$,

$$\|u\|_{C^{k,\gamma}(\Omega)} \leq c'_{k,m} \|u\|_{X'} .$$

By Leibniz' formula,

$$\begin{aligned} \|\partial^\alpha (uv)\|_{L^p(\Omega)} &\leq c_\alpha \|u\|_{C^{k,\gamma}(\Omega)} \|v\|_{C^{k,\gamma}(\Omega)} |\Omega|^{1/p} \\ &\leq c_\alpha (c'_{k,m})^2 |\Omega|^{1/p} \|u\|_{X'} \|v\|_{X'} , \qquad |\alpha| \leq k. \end{aligned}$$

Hence, after choosing the equivalent family of constants in the norm, there is $C'_1 > 0$ such that

$$\|uv\|_{X',\text{new}} \leq C'_1 \|u\|_{X',\text{old}} \|v\|_{X',\text{old}} .$$

From the proof of the Sobolev embedding theorem one may choose a constant C_4 independent of k so that, for every

$$\omega \Subset \Omega, \qquad \rho \in \mathcal{D}(\Omega), \qquad \rho|_\omega = 1,$$

$$\|\rho u\|_{C^k(\Omega)} \leq \|\rho u\|_{C^k(\mathbb{R}\times\mathbb{R}^3)} \leq C_4 \|\rho u\|_{W^{m,2}(\mathbb{R}\times\mathbb{R}^3)} \leq \frac{C_4}{\displaystyle\sum_{m\geq 5} \sum_{k \geq 0} \frac{1}{(m!)^4 (c'_{k,m})^2}} \|u\|_X .$$

Since ρ may be chosen arbitrarily, the corresponding constants $c''_{k,m}$ can be taken uniformly.

To compare the two choices of norm, choose positive constants satisfying

$$c'_{k,m} \geq 1, \qquad c''_{k,m} \geq c'_{k,m},$$

and a family

$$0 < \rho_{k,m} \leq 1$$

such that

$$\rho_{0,5} < \rho_{k+1,m+1} \leq \rho_{k,m},$$

and

$$0 < \frac{\rho_{k,m}}{(c'_{k,m})^2} \leq \frac{1}{(c''_{k,m})^2}.$$

Then the two resulting norms are equivalent. Consequently, after replacing the norm by an equivalent one if necessary, there exists $0 < \rho < 1$ such that

$$\rho \|uv\|_{X',\text{old}} \leq \|uv\|_{X',\text{new}} \leq C'_1 \|u\|_{X',\text{old}} \|v\|_{X',\text{old}} .$$

Taking

$$C_1 = \frac{C'_1}{\rho}$$

gives the assertion. $\square$

Lemma 3.7 (Absorption of differential)**.** *There exists $C_2 > 0$ such that*

$$\|\partial_{x^j} u\|_X \leq C_2 \|u\|_X$$

for every $u \in S$ and $j = 1, 2, 3$.

Lemma 3.8 (Convolution estimate)**.** *There exists $C_3 > 0$ such that, for every*

$$u \in S \setminus \{v \in S : \partial^\alpha v = 0 \text{ for some } \alpha\},$$

one has

$$E * (\chi_\Omega u) \in X$$

and

$$\|E * (\chi_\Omega u)\|_X \leq C_3 \|u\|_X,$$

together with

$$\|E * \partial_{x^j}(\chi_\Omega u)\|_X \leq C_3 C_2 \|u\|_X.$$

Proof. For almost every fixed $(t, x) \in \Omega$, viewed as a function of (s, y),

$$\operatorname{supp}(E(s, y)(\chi_\Omega u)(t - s, x - y)) \subset -\Omega + (t, x),$$

and the latter set is compact.

For every multi-index α,

$$\left|\partial_{t,x}^\alpha [E(s, y)(\chi_\Omega u)(t - s, x - y)]\right| \leq E(s, y) \sup_\Omega |\partial^\alpha u(t - s, x - y)|.$$

The right-hand side is integrable in (s, y) on the translated compact domain. Hence differentiation under the integral sign is justified.

Combining differentiation under the integral sign, Hölder's inequality, and the continuous embedding

$$S \hookrightarrow L^\infty(\Omega),$$

we obtain

$$\begin{aligned} \|\partial^\alpha(E * (\chi_\Omega u))\|_{L^p(\Omega)} &\leq \sup_{(t,x)\in\Omega} \|E\|_{L^1(-\Omega+(t,x))} |\Omega|^{1/p} \|\partial^\alpha u\|_{L^\infty(\Omega)} \\ &\leq C_\alpha |\Omega|^{1/p} \|u\|_X. \end{aligned}$$

After summing with the defining weights of X,

$$\|E * (\chi_\Omega u)\|_X \leq C_3 \|u\|_X.$$

Applying the same argument after one spatial differentiation and using the previous lemma yields

$$\|E * \partial_{x^j}(\chi_\Omega u)\|_X \leq C_3 C_2 \|u\|_X.$$

The derivative is understood through the common derivative almost everywhere on Ω; no derivative of the characteristic function is used in the interior estimate. □

Take

$$C = \max\{C_1, C_2, C_3\}.$$

The constants in the preceding estimates may be arranged so that

$$C = O(|\Omega|).$$

We choose Ω, f, M so that
$$C\,\|f\|_{X'} + 3C^3M^2 \le M.$$

We solve the projected equation
$$\partial_t u - \Delta u = P\big(f - (u\cdot\nabla)u\big). \tag{N-S$'$}$$

Define
$$\Phi : S \longrightarrow X$$
by
$$\Phi[u](t,x) = \int_{\mathbb{R}\times\mathbb{R}^3} E(s,y)\big(\chi_\Omega P(f-(u\cdot\nabla)u)\big)(t-s,x-y)\,ds\,dy.$$

Choose
$$u_0 \in S, \qquad u_1 \in S, \qquad u_1 - u_0 \in S,$$
and for $n \ge 1$ put
$$u_{n+1} = \Phi[u_n].$$

Lemma 3.9 (Well-definedness of Φ)**.** *For every $u \in S$,*
$$\|E * (\chi_\Omega P(f-(u\cdot\nabla)u))\|_X < \infty.$$

Proof. Since
$$\|P\| \le 1,$$
the preceding lemmas give
$$\begin{aligned}\|E * (\chi_\Omega P(f-(u\cdot\nabla)u))\|_X &\le C\,\|f\|_{X'} + C\,\big\|u^1\partial_{x_1}u + u^2\partial_{x_2}u + u^3\partial_{x_3}u\big\|_{X'}\\ &\le C\,\|f\|_{X'} + 3C^3M^2 < \infty.\end{aligned}$$
By the choice of Ω, f, M,
$$C\,\|f\|_{X'} + 3C^3M^2 \le M,$$
and therefore
$$\boxed{\|\Phi[u]\|_X \le M} \qquad (u\in S).$$
$\square$

Lemma 3.10 (The iterates remain in S)**.** *Let $u_n \in S$ and put $u_{n+1} = \Phi[u_n]$. Then*
$$u_{n+1} \in S.$$
Consequently, if $u_0 \in S$, all the iterates constructed by $u_{n+1} = \Phi[u_n]$ belong to S.

Proof. Lemma 3.9 and the smallness condition give
$$\|\Phi[u_n]\|_X \le M.$$
Thus the norm condition in the definition of S is already satisfied. In fact, for the construction of the solution, the second inequality in Lemma 3.8 alone is sufficient once the uniform bound $\|\Phi[u_n]\|_X \le M$ is available. We verify that a uniform derivative constant exists along the sequence of iterates. Suppose, to the contrary, that no such constant exists. Since there are only three spatial directions, after passing to a subsequence we may fix $j \in \{1,2,3\}$ and choose numbers $C_n \to \infty$ such that
$$C_n\|\Phi[u_n]\|_X < \|\partial_{x_j}\Phi[u_n]\|_X.$$
The contraction argument gives $u_n \to u$ in X, and hence $\Phi[u_n] = u_{n+1} \to u$ in X. Under the non-degeneracy assumption on the force in the Assumption above, $f \ne 0$ implies that the fixed

point is nonzero: indeed, $u = 0$ in the fixed-point equation would give $E * (\chi_\Omega P f) = 0$, contrary to that assumption. Consequently

$$\|\Phi[u_n]\|_X \longrightarrow \|u\|_X > 0,$$

so there are $m > 0$ and N such that

$$\|\Phi[u_n]\|_X \geq m \qquad (n \geq N).$$

On the other hand, Lemmas 3.6–3.8 and $\|u_n\|_X \leq M$ give a constant $K < \infty$, independent of n, such that

$$\|\partial_{x^j} \Phi[u_n]\|_X \leq K.$$

Indeed, the differentiated convolution is estimated by the same product and convolution bounds used in Lemma 3.9, with one spatial derivative. Therefore, for $n \geq N$,

$$C_n m \leq C_n \|\Phi[u_n]\|_X < \|\partial_{x^j} \Phi[u_n]\|_X \leq K,$$

which is impossible as $C_n \to \infty$. Hence a finite constant $C_2 > 0$ exists such that

$$\|\partial_{x^j} \Phi[u_n]\|_X \leq C_2 \|\Phi[u_n]\|_X , \qquad j = 1, 2, 3,$$

for all iterates. Enlarging the constant C in the definition of S, if necessary, so that $C \geq C_2$, we obtain $\Phi[u_n] \in S$. Since $u_{n+1} = \Phi[u_n]$, induction gives

$$\boxed{u_n \in S \quad \text{for every } n.}$$

Thus the required invariance is verified along the actual sequence of iterates, rather than assumed a priori. □

Lemma 3.11 (Φ is a contraction)**.** *There exists a constant $0 < L < 1$ such that*

$$\|\Phi[v] - \Phi[u]\|_X \leq L \|u - v\|_X$$

whenever $u, v \in S$ and the indicated differences belong to the class on which the derivative estimates are available.

Proof. We use

$$(v \cdot \nabla)v - (u \cdot \nabla)u = \sum_{j=1}^{3} \left[v^j \partial_{x^j}(v - u) + (v^j - u^j)\partial_{x^j} u \right].$$

Hence

$$\begin{aligned} &\left\| \int_{\mathbb{R}\times\mathbb{R}^3} E(s, y) \left[\chi_\Omega P \left((v \cdot \nabla)v - (u \cdot \nabla)u\right)\right](t - s, x - y)\, ds\, dy \right\|_X \\ &\quad \leq C^2 \|v\|_X \max_j \|\partial_{x^j}(v - u)\|_X + C^2 \|v - u\|_X \max_j \|\partial_{x^j} u\|_X \\ &\quad \leq 2C^3 M \|u - v\|_X . \end{aligned}$$

Therefore one may take

$$L = 2C^3 M.$$

Choose

$$\|f\|_{X'} \leq \frac{1}{12C^4}.$$

From

$$C \|f\|_{X'} + 3C^3 M^2 \leq M,$$

one may choose M so that

$$2C^3 M \leq \frac{1 - \sqrt{1 - 12C^4 \|f\|_{X'}}}{3} < 1.$$

Thus $0 < L < 1$. □

Fixed-point theorem used in the construction

Theorem 3.12 (Fixed-point theorem)**.** *Let S be a closed subset of a Banach space X, and let*

$$\Phi : S \longrightarrow X.$$

Assume that there exists $0 < L < 1$ such that

$$\|\Phi[u] - \Phi[v]\|_X \leq L \|u - v\|_X$$

for every relevant pair $u, v \in S$.

Suppose that a sequence can be defined by

$$u_{n+1} = \Phi[u_n], \qquad u_n \in S$$

for every n. Then $\{u_n\}$ converges in X to a point $u \in S$, and

$$u = \Phi[u].$$

In particular, the assumption

$$\Phi[S] \subset S$$

is not required once such an iterative sequence in S has been constructed.

If

$$u = \Phi[u], \qquad v = \Phi[v],$$

then $u = v$.

Proof. Since

$$u_{n+1} = \Phi[u_n],$$

we have

$$\|u_{n+1} - u_n\|_X \leq L \|u_n - u_{n-1}\|_X .$$

Hence, by induction,

$$\|u_{n+1} - u_n\|_X \leq L^{n-1} \|u_2 - u_1\|_X , \qquad n \geq 2.$$

For $m > n$,

$$\begin{aligned} \|u_m - u_n\|_X &\leq \sum_{j=n}^{m-1} \|u_{j+1} - u_j\|_X \\ &\leq \frac{L^{n-1}}{1-L} \|u_2 - u_1\|_X . \end{aligned}$$

Thus $\{u_n\}$ is Cauchy. Since X is complete, there exists $u \in X$ such that

$$u_n \longrightarrow u \qquad \text{in } X.$$

Because S is closed and $u_n \in S$,

$$u \in S.$$

By Lipschitz continuity,

$$\Phi[u_n] \longrightarrow \Phi[u].$$

Since

$$\Phi[u_n] = u_{n+1} \longrightarrow u,$$

we obtain

$$u = \Phi[u].$$

Finally,

$$\|u - v\|_X = \|\Phi[u] - \Phi[v]\|_X \leq L \|u - v\|_X ,$$

and $L < 1$ implies $u = v$. □

For the present construction, the preceding lemma shows directly that

$$u_n \in S \qquad \text{for every } n.$$

Moreover, the derivative estimates give

$$\|\partial_{x^j}(u_{n+1} - u_n)\|_X \le C \|u_{n+1} - u_n\|_X .$$

Hence

$$\|u_{n+1} - u_n\|_X \le L^{n-1} \|u_2 - u_1\|_X , \qquad n \ge 2.$$

Thus the hypotheses required for the fixed-point construction are verified along the sequence of iterates itself.

Lemma 3.13. *Let $U \in X$. Then*

$$\langle U, \varphi\rangle = 0$$

for every $\varphi \in \mathcal{D}_\sigma(\Omega)$ if and only if there exists a distribution p such that

$$U = \nabla p.$$

Proof. If $U = \nabla p$, then for every $\varphi \in \mathcal{D}_\sigma(\Omega)$,

$$\langle \nabla p, \varphi\rangle = -\int_\Omega p \operatorname{div} \varphi \, dt\, dx = 0.$$

Conversely, by the Helmholtz decomposition, if

$$U = PU + \nabla p$$

and U annihilates every divergence-free test function, its divergence-free part vanishes, and hence

$$U = \nabla p.$$

□

Lemma 3.14 (Solvability of the Navier–Stokes equations)**.** *The fixed point u of Φ is a solution of* (N−S′)*, and there exists p such that (u, p) solves the Navier–Stokes equation in the distributional sense.*

Proof. For every $u \in W^{m,p}_\sigma(\Omega)$, there is a Cauchy sequence

$$u_n \in V^{m,p}_\sigma(\Omega)$$

converging to u. By integration by parts and Hölder's inequality,

$$0 = -\sum_{j=1}^{3} \left\langle u_n^j, \partial_{x^j}\varphi\right\rangle \longrightarrow -\sum_{j=1}^{3} \left\langle u^j, \partial_{x^j}\varphi\right\rangle ,$$

hence

$$\operatorname{div} u = 0$$

in $\mathcal{D}'(\Omega)$.

The Sobolev embedding theorem gives boundedness of u and its first spatial derivatives. Since $|\Omega| < \infty$,

$$(u \cdot \nabla)u \in L^2(\Omega).$$

Write the Helmholtz decompositions

$$f = Pf + \nabla f_Q, \qquad (u \cdot \nabla)u = P((u \cdot \nabla)u) + \nabla u_Q.$$

For $\varphi \in \mathcal{D}_\sigma(\Omega)$,
$$\langle f, \varphi\rangle = \langle Pf, \varphi\rangle\,, \qquad \langle (u\cdot\nabla)u, \varphi\rangle = \langle P((u\cdot\nabla)u), \varphi\rangle\,.$$

For the approximate equation
$$\partial_t v_n - \Delta v_n = P(f - (u_n\cdot\nabla)u_n),$$
Proposition 1 gives
$$v_n = u_{n+1} = E * (\chi_\Omega P(f - (u_n\cdot\nabla)u_n))\,.$$
Therefore
$$(\partial_t - \Delta)u_{n+1} = P(f - (u_n\cdot\nabla)u_n).$$

By Hölder's inequality, $\|P\| \leq 1$, and continuity of
$$L^2(\Omega)\times L^2(\Omega) \longrightarrow L^1(\Omega), \qquad (u,v)\longmapsto uv,$$
we obtain
$$\left|\int_\Omega P\left((u_n\cdot\nabla)u_n - (u\cdot\nabla)u\right)\cdot\varphi\right| \longrightarrow 0.$$
By continuity of the heat operator on distributions,
$$\langle (\partial_t - \Delta)u_n, \varphi\rangle \longrightarrow \langle (\partial_t - \Delta)u, \varphi\rangle\,.$$
Hence
$$(\partial_t - \Delta)u = P(f - (u\cdot\nabla)u)$$
in $\mathcal{D}'(\Omega)$.

Since
$$u = \Phi[u],$$
$$u(t,x) = \int_{\mathbb{R}\times\mathbb{R}^3} E(s,y)\left(\chi_\Omega P(f - (u\cdot\nabla)u)\right)(t-s, x-y)\,ds\,dy.$$
By the preceding Helmholtz lemma, there exists p such that
$$(\partial_t - \Delta)u + (u\cdot\nabla)u - f = -\nabla p.$$

□

4 Properties of the solutions

Lemma 4.1 (Vanishing)**.** *For every multi-index α,*
$$\lim_{t,\,|x|\to\infty} \partial^\alpha u(t,x) = 0.$$

Proof. From the integral representation,
$$\partial^\alpha u(t,x) = \int_\Omega E(t-s, x-y)\partial^\alpha P(f - (u\cdot\nabla)u)(s,y)\,ds\,dy.$$
For $t - s > t_0 > 0$,
$$|E(t-s, x-y)| \leq \frac{C}{(t-s)^{3/2}},$$
while
$$\partial^\alpha P(f - (u\cdot\nabla)u)$$
is bounded on the bounded domain Ω. The assertion follows from the bounded convergence theorem. □

The domain Ω may be taken arbitrarily large, so this vanishing property may be interpreted as a boundary condition at infinity.

Lemma 4.2 (Continuity of $f \mapsto u$)**.** *Let f_n, f satisfy*

$$Pf_n, Pf \in S, \qquad \|f_n - f\|_{X'} \longrightarrow 0.$$

Let u_n and u be the corresponding solutions. Then

$$\|u_n - u\|_X \longrightarrow 0.$$

Proof. The contraction estimate gives

$$\|u_n - u\|_X \leq C \|f_n - f\|_{X'} + 2C^3 M \|u_n - u\|_X .$$

Thus

$$\limsup_{n\to\infty} \|u_n - u\|_X \leq 2C^3 M \limsup_{n\to\infty} \|u_n - u\|_X .$$

Since

$$2C^3 M < 1,$$

it follows that

$$\lim_{n\to\infty} \|u_n - u\|_X = 0.$$

□

Proposition 4.3 (Elementary weak solutions as elements of a Bochner class)**.** *Let*

$$I = \left\{ t \in \mathbb{R} : \exists x \in \mathbb{R}^3,\ (t,x) \in \Omega \right\},$$

and

$$\Omega' = \left\{ x \in \mathbb{R}^3 : \exists t \in \mathbb{R},\ (t,x) \in \Omega \right\}.$$

For $1 \leq p < \infty$,

$$a \in C^\infty(\Omega') \cap L^p_\sigma(\Omega'), \qquad u \in C(I; L^p_\sigma(\Omega')).$$

Proof. Since

$$u \in C^{0,\alpha}(\Omega),$$

for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|(t,x) - (t',x)| < \delta$$

implies

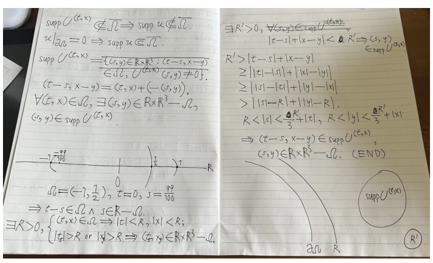
$$|u(t,x) - u(t',x)| < \varepsilon.$$

Therefore

$$\left\| u(t,\cdot) - u(t',\cdot) \right\|_{L^p(\Omega')} \leq |\Omega'|^{1/p} \varepsilon.$$

Hence

$$u \in C(I; L^p(\Omega')).$$



□

Lemma 4.4 (Boundary support observation)**.** *An elementary weak solution satisfies*

$$u(t,x) = \int_{\mathbb{R}\times\mathbb{R}^3} E(s,y)\,(\chi_\Omega P(f-(u\cdot\nabla)u))\,(t-s,x-y)\,ds\,dy.$$

For each (t,x), define

$$U_{(t,x)}(s,y) = (\chi_\Omega P(f-(u\cdot\nabla)u))\,(t-s,x-y).$$

If support were preserved in the naive sense

$$\operatorname{supp} U_{(t,x)} \subset \Omega \quad \Longleftrightarrow \quad \operatorname{supp} u \subset \Omega,$$

then integration by parts would immediately yield the energy inequality. However, this support inclusion does not hold in general.

Indeed, since Ω is bounded, there is $R > 0$ such that

$$(t,x) \in \Omega \quad \Longrightarrow \quad |t|, |x| < R.$$

If

$$(t_0, x_0) \in \operatorname{supp} U_{(t,x)} \cap \Omega,$$

then there is $R' > 0$ such that

$$|t_0 - s| + |x_0 - y| < R' \quad \Longrightarrow \quad (s,y) \in \operatorname{supp} U_{(t,x)}.$$

But

$$|t_0 - s| + |x_0 - y| \geq \big||s| - |t_0|\big| + \big||y| - |x_0|\big|.$$

For sufficiently large $|s|$ and $|y|$, the right-hand side is larger than R', while $(s,y) \notin \Omega$. Therefore the above support inclusion cannot be used as a general boundary argument.

Proposition 4.5 (Analyticity)**.** *The elementary weak solution u is analytic.*

Proof. The constants in the Sobolev embedding estimates may be chosen uniformly with respect to the differentiation order. Consequently, there exists a constant $C > 0$ such that for every multi-index

$$\alpha \in \mathbb{Z}^4_{\geq 0},$$

$$\sup_{(t,x)\in\Omega} |\partial^\alpha u(t,x)| \leq C.$$

Let

$$z = (t,x) \in \Omega$$

and let $h \in \mathbb{R}^4$ be small enough that the segment from z to $z+h$ lies in Ω. Taylor's formula gives

$$u(z+h) = \sum_{|\alpha|\leq N} \frac{\partial^\alpha u(z)}{\alpha!} h^\alpha + R_N(z,h).$$

The uniform derivative bound implies

$$|R_N(z,h)| \leq C \sum_{|\alpha|=N+1} \frac{|h^\alpha|}{\alpha!} \longrightarrow 0 \qquad (N \to \infty).$$

Hence the Taylor series converges locally to u, and u is analytic. $\square$

Proposition 4.6 (Energy inequality). *Let u be the extended elementary weak solution on $\mathbb{R}\times\mathbb{R}^3$ given by the convolution representation. Then, for $t \geq 0$,*

$$\frac{1}{2}\left\|u(t)\right\|_{L^2(\mathbb{R}^3)}^2 + \int_0^t \left\|\nabla u(s)\right\|_{L^2(\mathbb{R}^3)}^2 \, ds \leq \frac{1}{2}\left\|u(0)\right\|_{L^2(\mathbb{R}^3)}^2 + \int_0^t (f(s), u(s))_{L^2(\mathbb{R}^3)} \, ds.$$

For the smooth solutions considered here, the corresponding energy identity holds whenever all terms are finite.

Proof. We cut off the integration at spatial infinity and then use Lemma 4.1. Choose $\eta \in C_0^\infty(\mathbb{R}^3)$ such that

$$0 \leq \eta \leq 1, \qquad \eta = 1 \text{ on } B_1(0), \qquad \eta = 0 \text{ outside } B_2(0),$$

and put $\eta_R(x) = \eta(x/R)$. Then

$$|\nabla \eta_R| \leq \frac{C}{R}, \qquad \operatorname{supp} \nabla\eta_R \subset A_R := \{x : R < |x| < 2R\}.$$

Multiply the Navier–Stokes equation by $\eta_R^2 u$ and integrate over $\mathbb{R}^3$. For the viscous term, integration by parts gives

$$\begin{aligned}-\int_{\mathbb{R}^3} \Delta u \cdot (\eta_R^2 u)\, dx &= \int_{\mathbb{R}^3} \eta_R^2 |\nabla u|^2 \, dx \\ &\quad + 2\sum_{i,j=1}^3 \int_{\mathbb{R}^3} \eta_R u^i \partial_{x^j} u^i \partial_{x^j} \eta_R \, dx.\end{aligned}$$

The last integral is supported in A_R, and

$$\left| 2\sum_{i,j=1}^3 \int \eta_R u^i \partial_{x^j} u^i \partial_{x^j}\eta_R \, dx \right| \leq \frac{C}{R} \left\|u\right\|_{L^2(A_R)} \left\|\nabla u\right\|_{L^2(A_R)} \longrightarrow 0$$

as $R \to \infty$. Thus no boundary condition on a finite sphere is needed; the boundary term is cut off at infinity.

Since $\operatorname{div} u = 0$, the nonlinear term satisfies

$$\begin{aligned}\int_{\mathbb{R}^3} ((u\cdot\nabla)u)\cdot u\, \eta_R^2 \, dx &= \frac{1}{2}\int_{\mathbb{R}^3} u \cdot \nabla |u|^2 \, \eta_R^2 \, dx \\ &= -\frac{1}{2}\int_{\mathbb{R}^3} |u|^2 u \cdot \nabla(\eta_R^2)\, dx.\end{aligned}$$

Again the right-hand side is supported in A_R. By Lemma 4.1, together with the Sobolev bounds defining X, the functions u and their spatial derivatives vanish at infinity and the corresponding tail terms tend to zero. Consequently,

$$\int_{\mathbb{R}^3} ((u\cdot\nabla)u)\cdot u \, dx = 0.$$

Similarly, after integration by parts,

$$\int_{\mathbb{R}^3} \nabla p \cdot (\eta_R^2 u)\, dx = -\int_{\mathbb{R}^3} p\, u \cdot \nabla(\eta_R^2)\, dx,$$

and the cut-off term tends to zero as $R \to \infty$; equivalently, one may use the projected equation and the fact that the gradient part is orthogonal to divergence-free fields.

Letting $R \to \infty$ therefore gives

$$\frac{1}{2}\frac{d}{dt}\left\|u(t)\right\|_{L^2(\mathbb{R}^3)}^2 + \left\|\nabla u(t)\right\|_{L^2(\mathbb{R}^3)}^2 = (f(t), u(t))_{L^2(\mathbb{R}^3)}.$$

Integrating from 0 to t yields the asserted energy inequality (indeed, the energy identity for the present smooth solution). □

The solutions are smooth and unique; u is defined on $\mathbb{R} \times \mathbb{R}^3$ through the convolution representation and satisfies

$$\lim_{t,\,|x| \to \infty} \partial^\alpha u(t,x) = 0$$

for every multi-index α. The map $f \mapsto u$ is continuous,

$$a \in C^\infty(\Omega') \cap L^p(\Omega'), \qquad u \in C(I; L^p(\Omega')).$$

One may alternatively strengthen the definition of S to

$$S = \left\{ u \in X : \|u\|_X \le M, \quad \left\| \partial_{x^j}^k u \right\|_X \le C^k \|u\|_X \,,\ j = 1,2,3,\ k \ge 0 \right\}.$$

With this definition the existence argument is unchanged. Each

$$\partial_{x^j} : X \longrightarrow W^{m,p}(\Omega)$$

is a closed operator, and the same is true for higher derivatives; hence S is closed by the same Fatou-type argument used in the proof of completeness.

For $f \neq 0$, many nonzero solutions can be constructed by varying the admissible norm constants. We also conjecture that blow-up solutions may be constructed as follows. For any $\varepsilon > 0$, there exist $\delta > 0$, a constant $M(\varepsilon) > 0$, and a force $f \in S_{M(\varepsilon)}$ such that

$$M(\varepsilon_1) > M(\varepsilon_2) \qquad \text{whenever } \varepsilon_1 > \varepsilon_2,$$

$$CM(\varepsilon) > \|E * (\chi_\Omega P f)\| > 2\varepsilon,$$

and

$$|\Omega| < \delta \quad \Longrightarrow \quad \|E * (\chi_\Omega P((u \cdot \nabla)u))\| < \varepsilon.$$

Then, by the triangle inequality,

$$\|u\| = \|\Phi[u]\| > \varepsilon.$$

We conjecture that large initial values may likewise be approached by shrinking $|\Omega|$.

5 Remarks

Remark 5.1. The same method applies to semilinear constant-coefficient equations

$$Lu = f(u)$$

on

$$S' = \left\{ u \in X' : \|u\|_{X'} \le M \right\}$$

provided that L has a locally integrable fundamental solution. Examples that we consider include the nonlinear heat equation, nonlinear Schrödinger equation, the three-dimensional nonlinear wave equation, and the KdV equation.

Remark 5.2. For divergence-free test functions it is enough to take

$$\psi \in \mathcal{D}(\Omega)$$

and set

$$\varphi = \operatorname{curl} \psi.$$

Remark 5.3. Suppose

$$\|u_n - u\|_{L^2(\Omega)} \longrightarrow 0, \qquad \|v_n - v\|_{L^2(\Omega)} \longrightarrow 0.$$

Then

$$\Big| \|u_n\|_{L^2(\Omega)} - \|u\|_{L^2(\Omega)} \Big| \le \|u_n - u\|_{L^2(\Omega)} .$$

For sufficiently large n,

$$\|u_n\|_{L^2(\Omega)} < \|u\|_{L^2(\Omega)} + 1.$$

Therefore

$$\begin{aligned} \|u_n v_n - uv\|_{L^1(\Omega)} &\le \|u_n\|_{L^2(\Omega)} \|v_n - v\|_{L^2(\Omega)} + \|v\|_{L^2(\Omega)} \|u_n - u\|_{L^2(\Omega)} \\ &< \Big(\|u\|_{L^2(\Omega)} + 1 \Big) \|v_n - v\|_{L^2(\Omega)} + \|v\|_{L^2(\Omega)} \|u_n - u\|_{L^2(\Omega)} \\ &\longrightarrow 0. \end{aligned}$$

Remark 5.4. We also consider a compressible-type extension. With modified assumptions and an initial density $\rho(0,\cdot)$, the system is written in the form

$$\begin{gathered} \partial_t \rho + \operatorname{div}(\rho u) = 0, \\ \operatorname{Div}\big(D(u) + \operatorname{div}(u)(\delta^{ij}) - p(\delta^{ij})\big) = f - \rho \left(\partial_t u + (u \cdot \nabla) u\right), \end{gathered}$$

where

$$(\operatorname{Div} T)^i = \sum_{j=1}^{3} \partial_{x^j} T^{ij}, \qquad D^{ij}(u) = \partial_{x^j} u^i + \partial_{x_i} u^j .$$

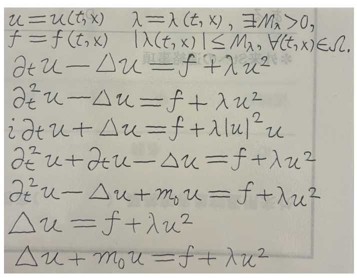
If the corresponding linear operator is bounded, one formally writes

$$\rho = e^{-tL_u} \rho(0,\cdot),$$

and substitutes this expression into the momentum equation. We states that the preceding arguments then apply similarly when the relevant elliptic fundamental solutions are locally integrable.

6 Extensions and applications

6.1 Extension as a solution and regularity



For

$$\varphi \in \mathcal{D}_\sigma(\mathbb{R} \times \mathbb{R}^3), \qquad \operatorname{supp} \varphi \subset \Omega,$$

one may extend the solution $\bar{u}$ to $\mathbb{R} \times \mathbb{R}^3$ so that

$$\bar{u} \in X(\mathbb{R} \times \mathbb{R}^3)$$

and

$$\partial_t \bar{u} - \Delta \bar{u} = Pf - P((\bar{u} \cdot \nabla)\bar{u})$$

in

$$\mathcal{D}'_\sigma(\mathbb{R} \times \mathbb{R}^3)$$

in the distributional sense appropriate to the construction.

The uniform Sobolev constants used above imply analyticity, as proved in Proposition 4.5.

6.2 Application to complex geometry and wave equations

We note that the elementary PDE theory can also be applied to two-dimensional wave equations, referring to Hörmander and the later references listed below. It further indicates an application to complex geometry through nonlinear elliptic equations with locally integrable fundamental solutions. The figures appearing in the source manuscript are omitted here.

We equip A with the metric

$$d_A(u(0), v(0)) = \|u - v\|_X .$$

With this topology, the map from initial values in A to the corresponding unique solutions is continuous.

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